Final Report

1. Introduction

The real estate market is influenced by various factors such as house age, renovation status, number of bedrooms and bathrooms, and overall size. This project aims to analyze housing market trends and visualize key insights using Tableau to better understand how different features impact sale prices.

1.1. Project overviews

The dataset contains Transformed housing data and 21,609 house sale records, including Property features such as Sales price, area, bedrooms, bathrooms, floors and location. There are a total of 31 columns, out of which Sale Price can be supposedly taken as a dependent variable. The other variables are different features, locations and date, etc. regarding the houses. This project, "Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau," aims to explore and analyze housing market trends using the Transformed Housing Data 2 dataset from Kaggle. The objective is to identify key factors influencing house prices, such as location, size, number of bedrooms, bathrooms, floors and basement area.

By leveraging Tableau, the project will create interactive dashboards, story, bar chart, histogram, summary dashboard to visualize patterns, compare regional price variations, and gain insights into how different features impact house sale prices. The analysis will help in making data-driven decisions for buyers, sellers, and real estate professionals.

1.2. Objectives

- Identify key factors influencing house prices.
- Analyze the effect of renovations on property value.
- Explore the distribution of house sales across different price ranges.
- Create interactive Tableau dashboards to present findings effectively.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A first-time homebuyer who wants to make an informed decision	Find a home within my budget that meets my needs	The available market data is difficult to interpret and scattered across multiple sources	There is no centralized, easy-to-use tool that visualizes housing trends based on historical sales data	Confused and overwhelmed, making me hesitant to proceed
PS-2	A real estate investor looking for high-return properties	Identify profitable properties based on price trends and key influencing factors	Existing datasets require extensive manual analysis and lack clear insights	No interactive visualization tool allows me to compare property appreciation trends effectively	Frustrated and uncertain about making investment decisions
PS-3	A real estate agent aiming to assist clients efficiently	Provide accurate and insightful recommendations based on market data	The data is time-consuming to analyze and spread across various reports	There is no comprehensive tool to aggregate and visualize pricing trends for quick insights	Less efficient, unable to provide quick, data- backed advice to clients

2.2 Empathy Map Canvas

2.3 Brain Storming

Step 1: Team Gathering, Collaboration and Problem Statement

Our team collaborated to identify pressing challenges in the real estate market, particularly in understanding how various property features influence housing sale prices. After exploring themes like housing affordability, real estate investment planning, urban development, and smart property

insights, we narrowed down our focus to uncover actionable insights hidden in housing data. The objective was to visually explore trends using Tableau that would help buyers, sellers, investors, and policy makers understand patterns of sale prices based on features like area, bedrooms, renovation status, condition, location (zipcode groups), and more.

Problem Statement:

4	Compare prices for renovated vs. non-renovated homes	Renovation Analysis
5	Group insights by Zipcode Clusters	Geographical Comparison
6	Analyze house condition vs. price using dummy variables	Quality-Based Pricing
7	Add calculated field: TotalAreaSqft	Data Preparation
8	Create SalePriceBin with 100k intervals	Binning / Categorization
9	Use Tableau dashboard to combine insights	Dashboard Design

10 Build a Story in Tableau for narrative Storytelling & Reporting

How can housing sale price trends and property characteristics be visualized and analyzed using Tableau to identify patterns, improve buyer/seller decision-making, and uncover insights that support strategic real estate planning?

Team Members:

- Team Leader: Teja paul
- Team Member: Singuluri charishma
- Team Member: seerapu Vishnu vilasini
- Team Member: NDSVSS Arthi

Step 2: Brainstorming, Idea Listing and Grouping

S.No	Idea Description	Category
1	Visualize average sale price by SalePriceBin	Pricing Insights
2	Analyze impact of number of bedrooms on sale price	Property Features
3	Explore relationship between Total Area and Price (scatter plot)	Size-Based Pricing
S.No	Idea Description	Category
11	Embed Dashboard in Web Application using Flask	Deployment & Integration
12	Add filters for Bedrooms, Condition, Renovation in Dashboard	Interactive Exploration

Step 3: Idea Prioritization Table

S.No	Idea Description	Impact	Feasibility	Priority
1	Visualize average sale price by SalePriceBin	High	Easy	High
2	Analyze impact of number of bedrooms on sale price	High	Easy	High
3	Explore TotalArea vs Price (scatter plot)	High	Easy	High
4	Compare prices for renovated vs. non-renovated homes	High	Medium	High
5	Group insights by Zipcode Clusters	Medium	Medium	Medium
6	Analyze house condition vs. price	High	Medium	High
7	Add calculated field: TotalAreaSqft	Medium	Easy	High

8	Create SalePriceBin with 100k intervals	Medium	Easy	High
9	Use Tableau dashboard to combine insights	High	Easy	High
10	Build a Story in Tableau	High	Medium	High
11	Embed Dashboard in Web Application	High	Hard	Medium
12	Add filters for Bedrooms, Condition, Renovation	Medium	Easy	Medium

3. Requirement analysis

3.1 Customer Journey map

Customer Journey Map: Housing Market Trends Dashboard

Stage	Touchpoints	Pain Points & Emotions	Opportunities	User Goals
Awareness	- Sees dashboard via social media, newsletter, Tableau Public - Reads title/summary	Unclear if dashboard is relevant Curious, Interested	Use benefit-driven titles, visual thumbnails purpose	Attract interest and clarify
	Actions &	**Experience**		

Consideration	- Clicks dashboard link - Reads introduction, explores layout - Uses filters for location, price, features - Views charts (bar, scatter, pie, etc) - Exports visuals - Shares dashboard - Bookmarks or downloads - Subscribes for updates - Revisits for new data - Leaves feedback	Engaged, Cautious Excited, Inquisitive Satisfied, Confident Loyal, Empowered	Overwhelmed by layout, unsure to start	Add guided walkthrough, simplify navigation	Understand the dashboard and its where features
Exploration			Filters not intuitive, charts slow to load	Add example queries, improve speed	Discover valuable insights

Decision	Limited export options or unclear findings formats	Enable easy download/share, offer export guides	Preserve and share
Retention	No update notifications, feedback unacknowledged	Enable email updates, actively respond to feedback	Stay informed and engaged

3.2 Solution Requirement

Functional Requirements (FRs)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**Data Import**	- Import data from CSV - Enable live database integration (MySQL)
FR-2	**Data Cleaning & Transformation**	- Handle missing values - Add calculated fields like Year, Lockdown
FR-3	**Data Visualization**	- Create Tableau worksheets - Build multiple dashboards - Enable filtering by region, year
FR-4	**User Interaction**	- View comparative bar charts - Analyze pre/post-lockdown trends
FR-5	**User Access**	- Role-based views for Analyst, Policy Maker, Developer - Download/export options
FR-6	**Feedback Loop**	- Allow stakeholder feedback and change requests - Implement revision cycles

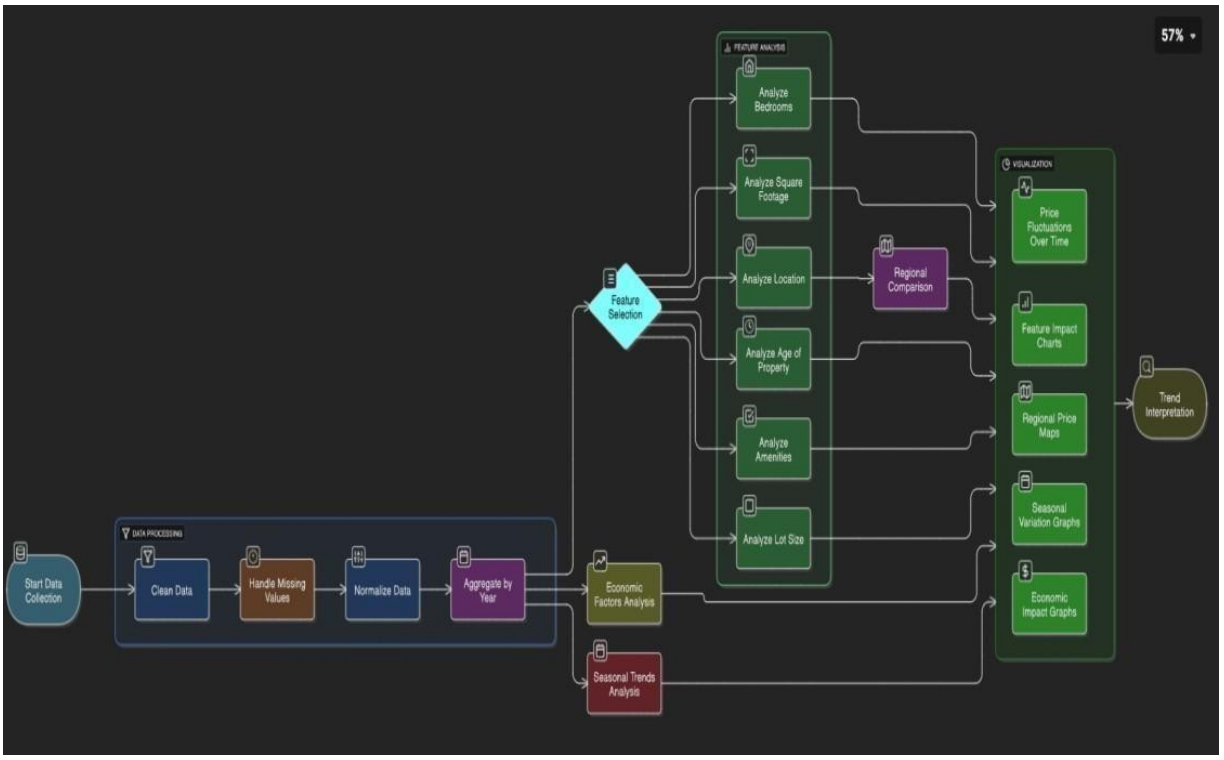
Non-Functional Requirements (NFRs)

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Dashboard must be intuitive with clear filters, legends, and guided walkthroughs
NFR-2	**Security**	Implement role-based access and secure backend/database connectivity
NFR-3	**Reliability**	System must handle unexpected data formats and maintain high accuracy
NFR-4	**Performance**	Ensure fast loading and responsive interaction across all dashboard elements
NFR-5	**Availability**	Dashboard should be accessible across browsers/devices with minimal downtime
NFR-6	**Scalability**	Should scale for large datasets and support additional features/modules

3.3 Data Flow Diagram

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

1. Data collected from POSOCO in CSV format.
2. Cleaned and transformed, with calculated fields like Year and Lockdown.
3. Visualizations built in Tableau using multiple worksheets.
4. Users review the dashboard and may request changes.
5. Final version archived after approval.

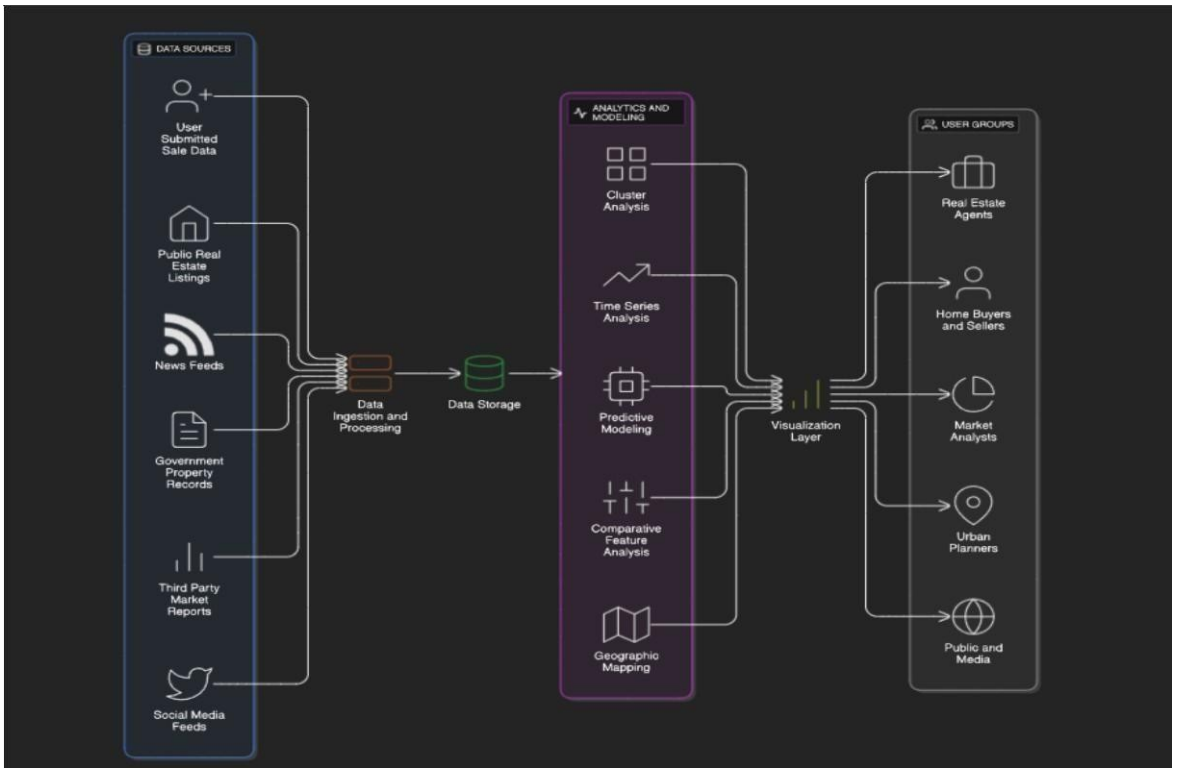


User Stories Table:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Analyst / Policymaker	View Electricity Trends	USN-1	As a user, I want to view electricity usage trends by specific regions and years.	I can filter and view charts for 1 region and year.	High	Sprint-1
			As a user, I want to view bar			

Analyst	Compare States	USN-2	compare top and bottom electricity-states.	charts with top N and bottom N 1	Medium consuming states.	Sprint-2
Policy Maker	Forecast Planning	USN-3	As a user, I want to analyze seasonal variations in usage.	I can view monthly and quarterly consumption trends.	High	Sprint-2
Energy Consultant	View Impact of Lockdown	USN-4	As a user, I want to compare electricity usage before and after lockdown.	A before-after lockdown chart is available for selection.	High	Sprint-2
Developer	Connect Data	USN-5	As a user, I want the dashboard to be automatically connected to a live database.	Data refreshes	Medium	Sprint-2
Developer	Export Insights	USN-6	As a user, I want to export dashboard views for presentations.	I can download dashboards as images or PDFs.	Low	Sprint-3

3.4 Technology Stack


DATA SOURCES
User Submitted Sale Data
Public Real Estate Listings
News Feeds
Government Property Records
Third Party Market Reports
Social Media Feeds
Data Ingestion and Processing
Data Storage
ANALYTICS AND MODELING
Cluster Analysis
Time Series Analysis
Predictive Modeling
Comparative Feature Analysis
Geographic Mapping
Visualization Layer
USER GROUPS
Real Estate Agents
Home Buyers and Sellers
Market Analysts
Urban Planners
Public and Media

4. Project design

4.1 Problem Solution Fit

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

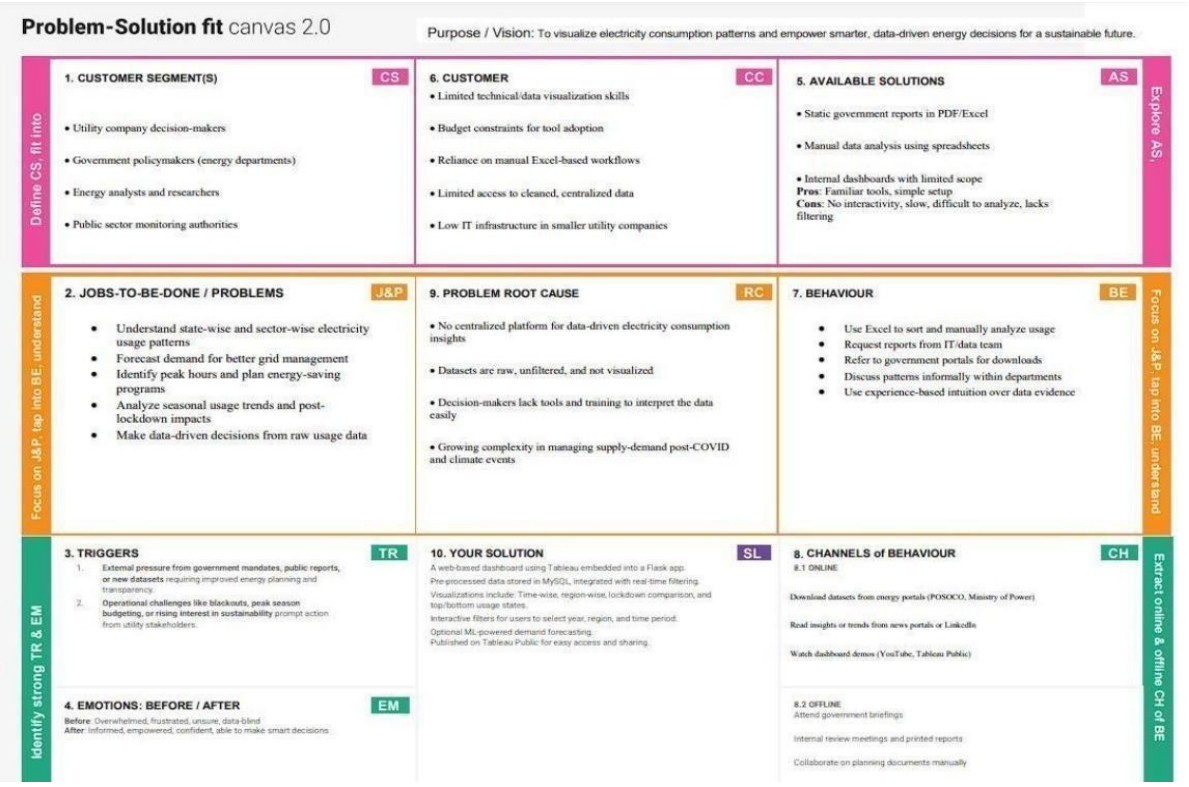
❑ Solve complex problems in a way that fits the state of your customers.

❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.

❑ Sharpen your communication and marketing strategy with the right triggers and messaging.

❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.

❑ Understand the existing situation in order to improve it for your target group.



4.2 Proposed Solution

Proposed Solution Template

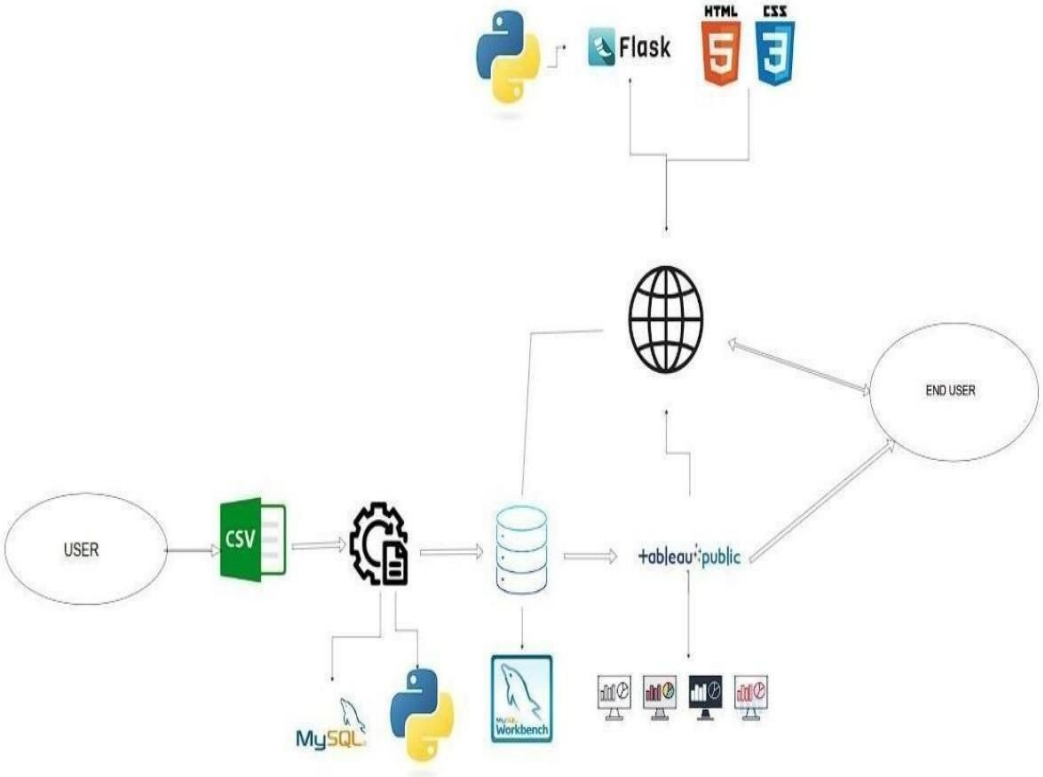
S.No.	Parameter	Description
1	**Problem Statement**	The real estate market involves vast and complex datasets on housing features and sale prices. These datasets are often underutilized due to lack of effective visualization, making it difficult for buyers, sellers, and analysts to draw insights or forecast trends.
2	**Idea / Solution Description**	Our solution transforms static housing datasets into interactive, insightful visualizations using Tableau. The project involves cleaning and transforming the data, creating calculated fields and KPIs, and developing a dashboard that highlights key trends, comparisons, and location-based analyses. The solution is deployed via a Flask web app.
3	**Novelty / Uniqueness**	This project leverages Tableau's powerful visual capabilities to go beyond basic data analytics. By combining calculated fields, condition segmentation, and geographic mapping, the dashboard offers a dynamic exploration of how features like bedrooms, area, renovation, and location influence housing prices.
4	**Social Impact / Customer Satisfaction**	This solution enables real estate buyers, sellers, agents, and market researchers to make informed decisions. It improves housing transparency, supports better urban planning, and enhances user engagement with clear visuals and actionable insights.
5	**Business Model (Revenue Model)**	This dashboard can be scaled and offered as a subscription-based SaaS tool to real estate companies, market research firms, or housing consultancies. Advanced forecasting modules, API integrations, and custom dashboards can be monetized as premium features.
6	**Scalability of the Solution**	The system is designed to be scalable and adaptable. It can incorporate new datasets (like rental trends or economic indicators), extend to new regions or cities, and integrate with ML models for price predictions, thereby offering long-term growth potential.

4.3 Solution Architecture

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- The architecture separates data preprocessing, storage, visualization, and UI layers—making it easy to maintain, scale, and enhance.

- Cleaned data from MySQL is visualized using Tableau dashboards, offering region-wise, year-wise, and seasonal insights with filtering capabilities.
- Dashboards are embedded into a Flask-based web interface, allowing end users to interact with visual data through a user-friendly portal.
- The solution supports future extensions like forecasting models and can be deployed locally or on cloud platforms like Heroku or AWS.



5. Project planning & scheduling

5.1 Project Planning

Sprint	Epic	User Story No.	User Story / Task	Points	Priority	Assigned To
Sprint-1	Data Setup	USN-1	As a user, I can upload housing data in CSV format	3	High	**N D SVSS arthi**
Sprint-1	Data Cleaning	USN-2	As a developer, I can clean and preprocess housing data in Tableau	4	High	**Teja paul**
Sprint-1	Field Creation	USN-3	As a user, I can create calculated fields like TotalAreaSqft	2	Medium	**N D SVSS arthi**

Sprint-1	Price Binning	USN-4	As a user, I can create SalePriceBin 2 for grouping houses	2	Medium	**Singuluri charishma**
Sprint-2	Data Visualization	USN-5	As a user, I can create sheets with 5 charts: price vs features	3	High	**Singuluri charishma**
Sprint-2	Dashboard Creation	USN-6	As a user, I can build an interactive Tableau Dashboard with filters	3	High	**N D SVSS arthi**
Sprint-2	Dashboard Styling	USN-7	As a user, I can style the dashboard for better readability and navigation	2	Medium	**Teja paul** **Teja paul**
Sprint-3	Storytelling	USN-8	As a user, I can create a Tableau Story showing insights step by step	2	Medium	**Seerapu Vishnu vilasini** **N D SVSS arthi**
Sprint-3	Flask Integration	USN-9	As a developer, I can embed Tableau dashboard into a Flask web app	4	High	**Seerapu Vishnu vilasini**
3	Embed Testing	USN-10	As a user, I can test and review the embedded dashboard UI	2	Medium	**Teja paul**
Sprint-4	Documentation	USN-11	As a team, we can prepare final project documentation	3	High	**N D SVSS arthi**
Sprint-4	Demo Preparation	USN-12	As a team, we can prepare and rehearse a full demo walkthrough	2	Medium	**Seerapu Vishnu vilasini** **Singuluri charishma**
Sprint-4	Bug Fixing / Final QA	USN-13	As a team, we can test the full system and fix visual/logic bugs	2	Medium	**N D SVSS arthi**

Project Tracker, Velocity & Burndown Chart

Sprin	**Total Story Poin**	**Duratio**	**Start Date**	**End Date**	**Points Complete**	**Release Da**
Sprint 11		4 Days	11 June 2021	14 June 2021	1	14 June 202
Sprint 10		4 Days	15 June 2021	18 June 2021	0	18 June 202
Sprint 7		4 Days	19 June 2022	22 June 2027		22 June 202
Sprint 7		4 Days	23 June 2022	26 June 2027		26 June 202

Velocity Calculation

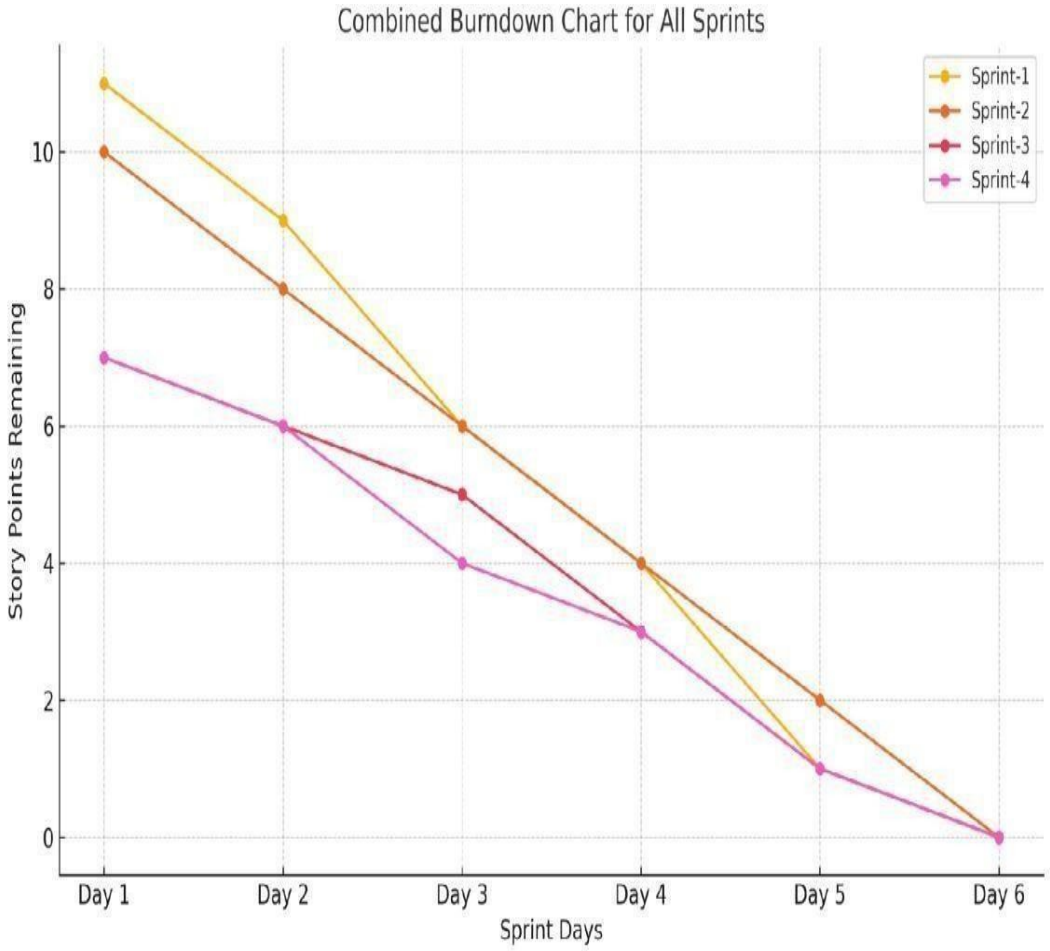
Total Points Completed: 11 + 10 + 7 + 7 = 35

Total Duration: 4 + 4 + 4 + 4 = 16 days

Average Velocity = Total Points Completed / Total Days = 35 / 16 = 2.19 points/day

Burndown Chart Insight

- Initial Total Story Points: 35
- Sprint-wise burn (Remaining Points): ○ After Sprint-1: 24 ○ After Sprint-2: 14 ○ After Sprint-3: 7 ○ After Sprint-4: 0



6. Functional and performance testing

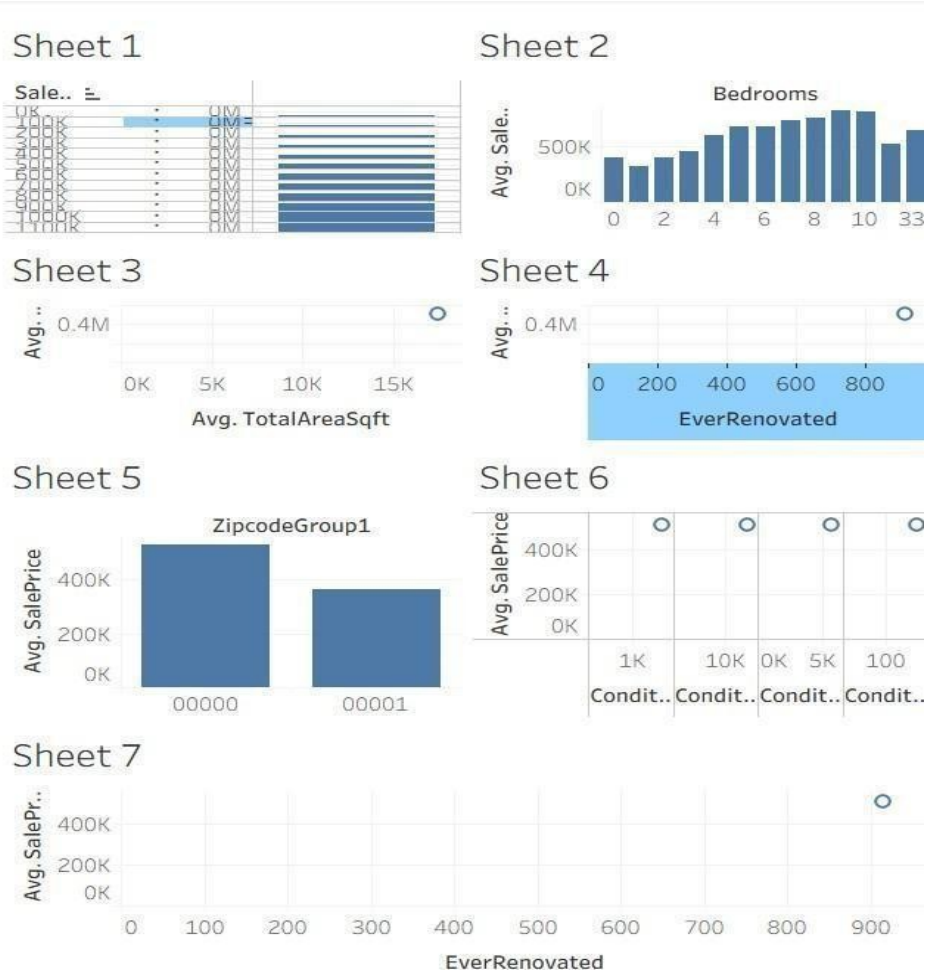
6.1 Performance Testing

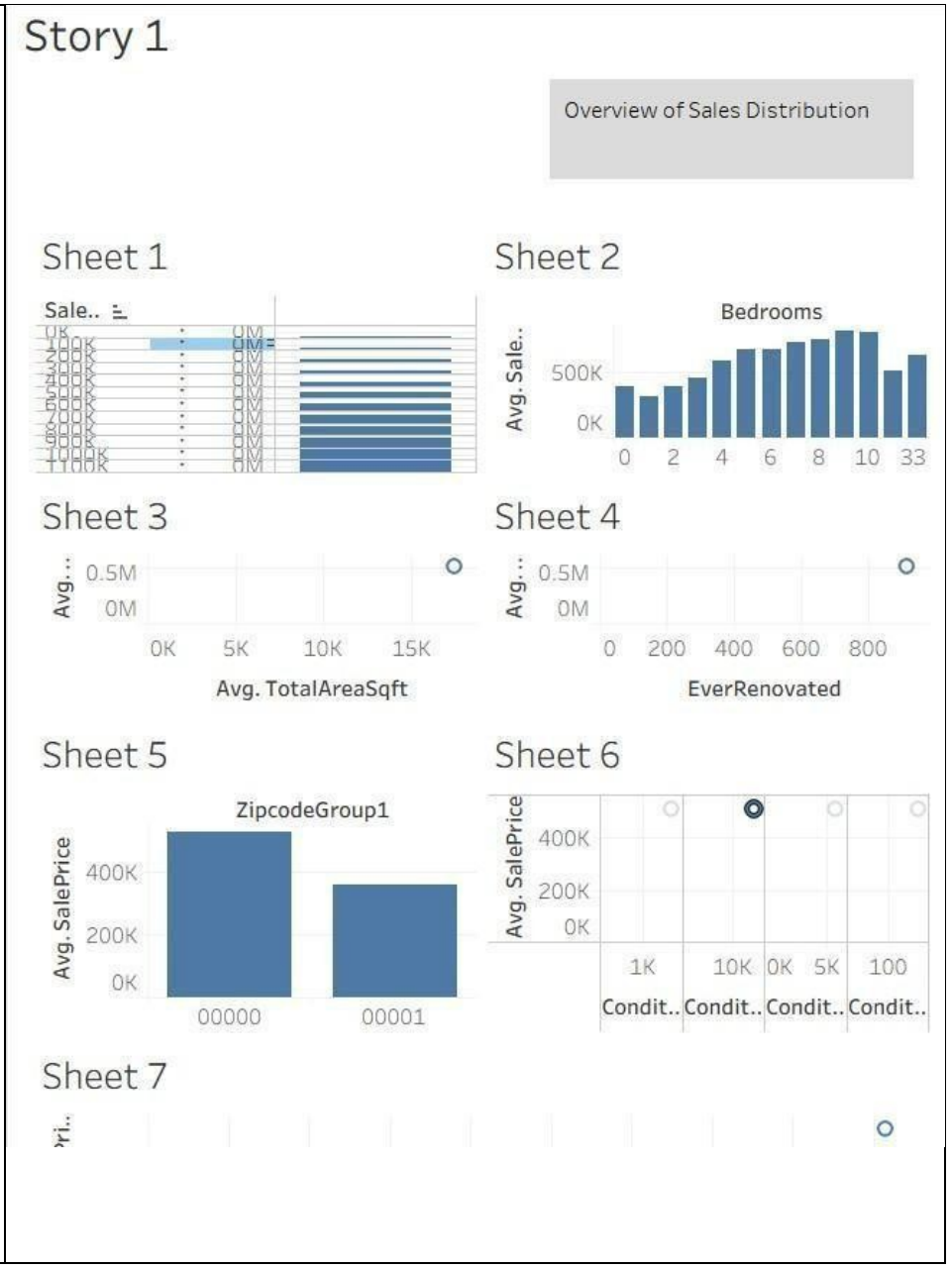
S.No	Parameter	Screenshot / Values

1.	**Data Rendered**	The dataset used contains housing sales data with fields such as Sale Price, Number of Bedrooms, Bathrooms, Flat Area, Lot Area, Basement Area, House Age, Condition, Renovation Status, Zipcode Group, and others. The data was provided in .csv format and include derived and transformed columns suitable for advanced analytics and visualizations in Tableau.
2.	**Data Preprocessin**	Before importing the data into Tableau, preprocessing was done usin Python (Pandas). The following steps were performed: • Removed null or missing values. • Renamed columns for clarity (e.g., "No of Bedrooms" → "Bedrooms"). • Created calculated fields like "TotalAreaSqft" (sum of flat, lo and basement areas). • Generated dummy variables for house conditions and renovation status. • Transformed categorical fields to improve Tableau usability. The final cleaned dataset was stored and imported into Tableau for visualization.

3.	**Utilization of Filters**	Multiple filters were implemented in Tableau to improve interactivity and user exploration. These include: • Number of Bedrooms • Number of Bathrooms • House Condition • Renovation Status (Yes/No) • Zipcode Group • Sale Price Bins These filters allow users to drill down and compare trends across different property types and regions.

4	**Calculated Fields Used**	Several calculated fields were created in Tableau to enhance analysis and interactivity: • TotalAreaSqft → [FlatAreaSqft] + [LotAreaSqft] + [BasementAreaSqft] • SalePriceBin → Binning Sale Price into ₹100,000 intervals • Condition_Excellent, Condition_Good, etc. → Dummy fields (0/1 • Ever_Renovated_Yes → Dummy field to identify renovated home • AvgPrice → AVG([SalePrice]) for grouped insights • HouseAge → Difference between year built and sale date if available (or derived field if pre-calculated) These fields enable comparisons across pricing, condition, and space utilization.

5	**Dashboard**	

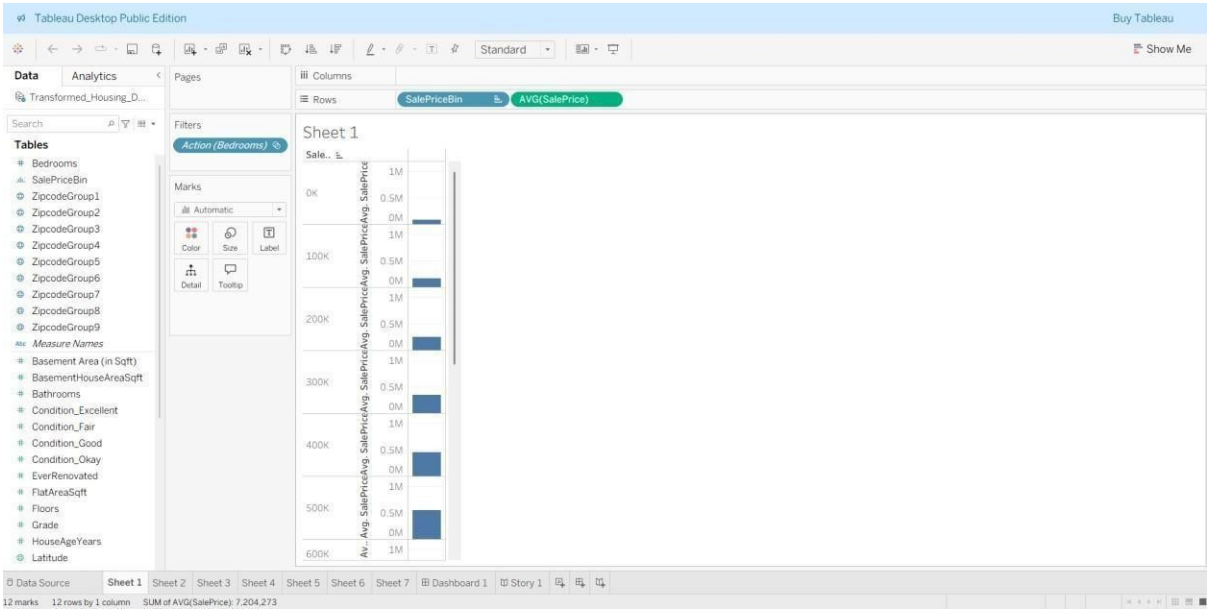
6	**Story Design**	

7. Results

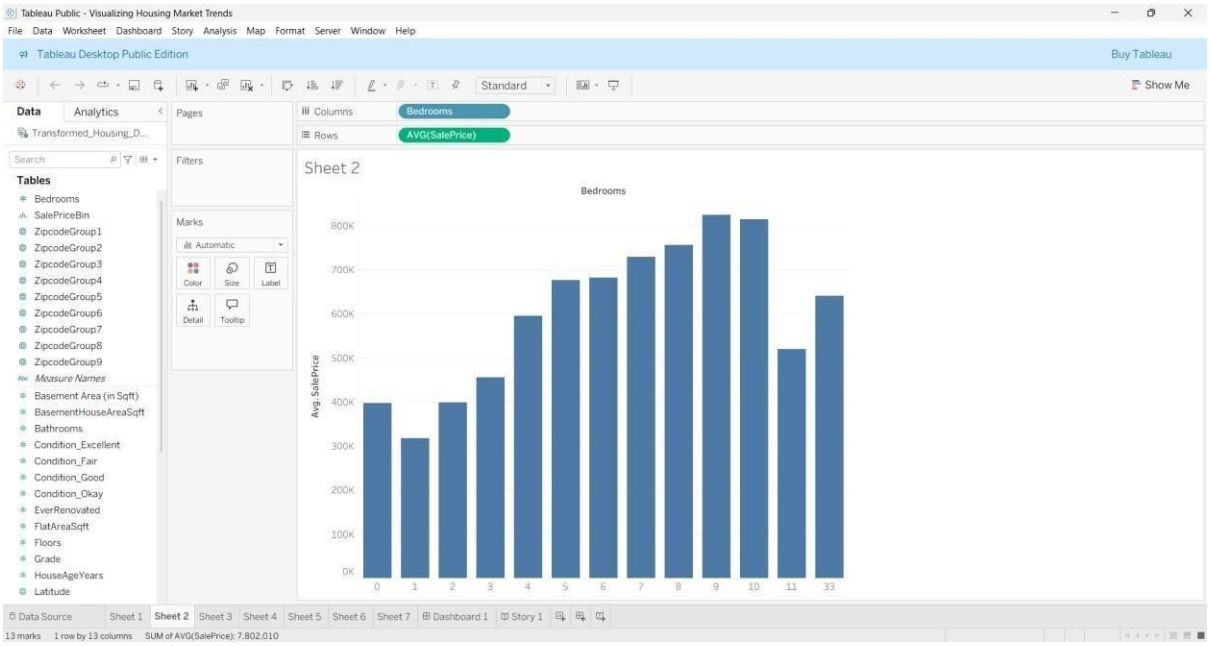
7.1 Output Screenshots

Output of Sheet 1

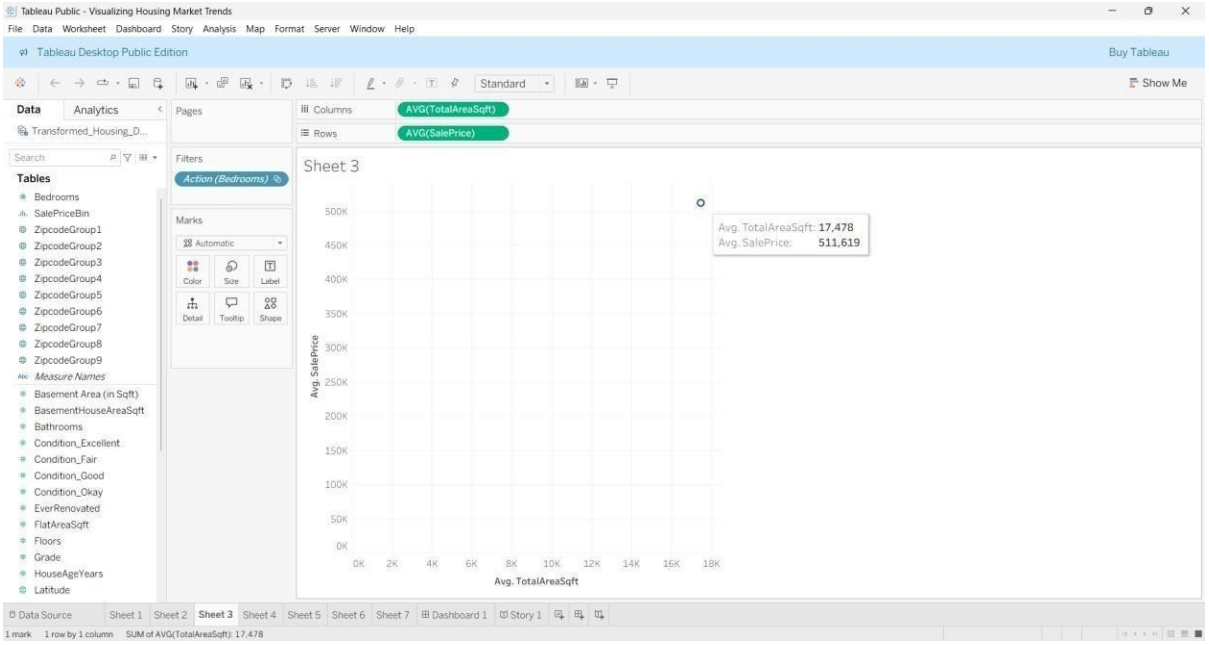
Output of



Output of Sheet 2



Sheet 3



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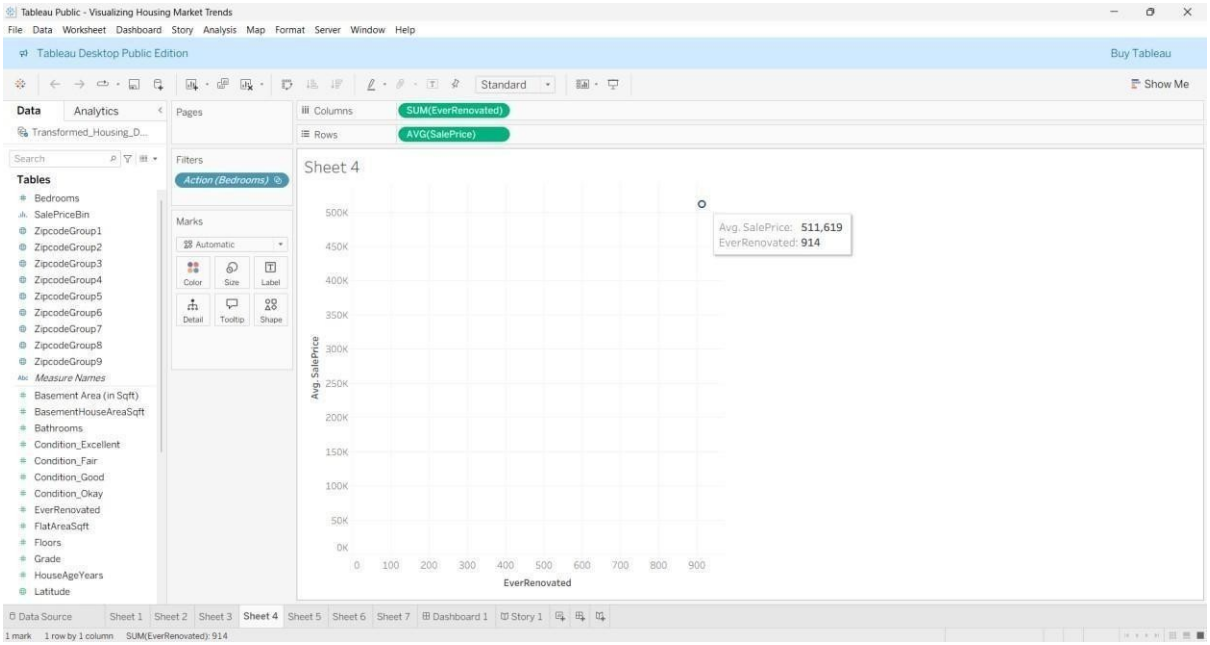
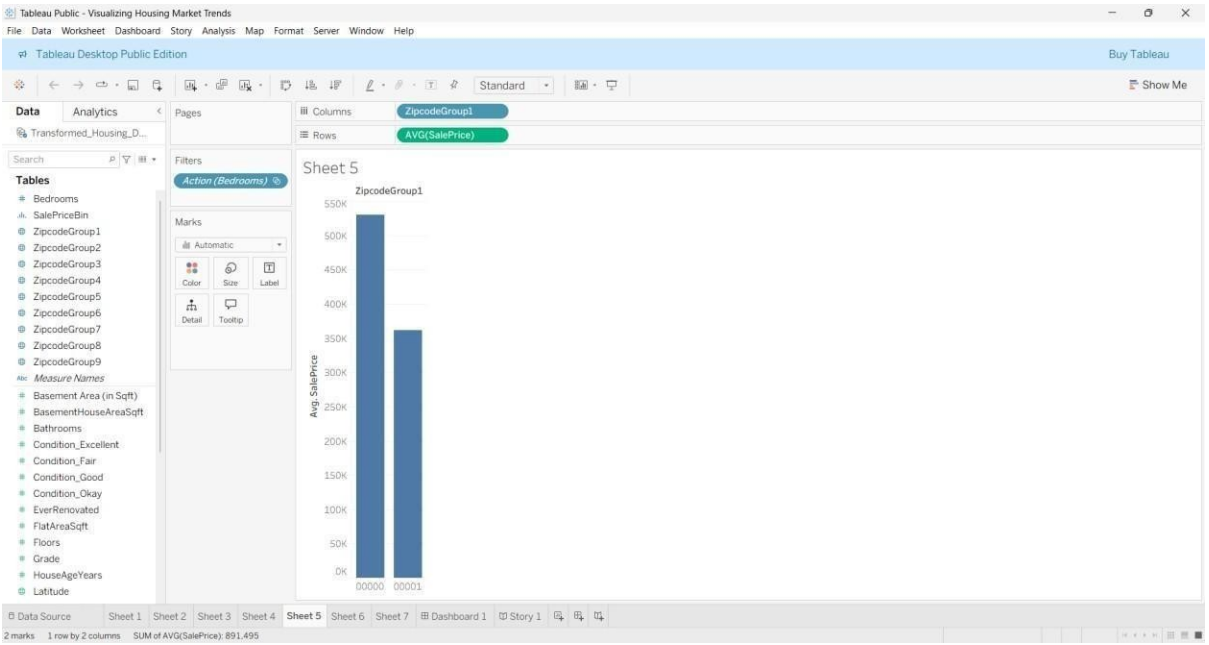


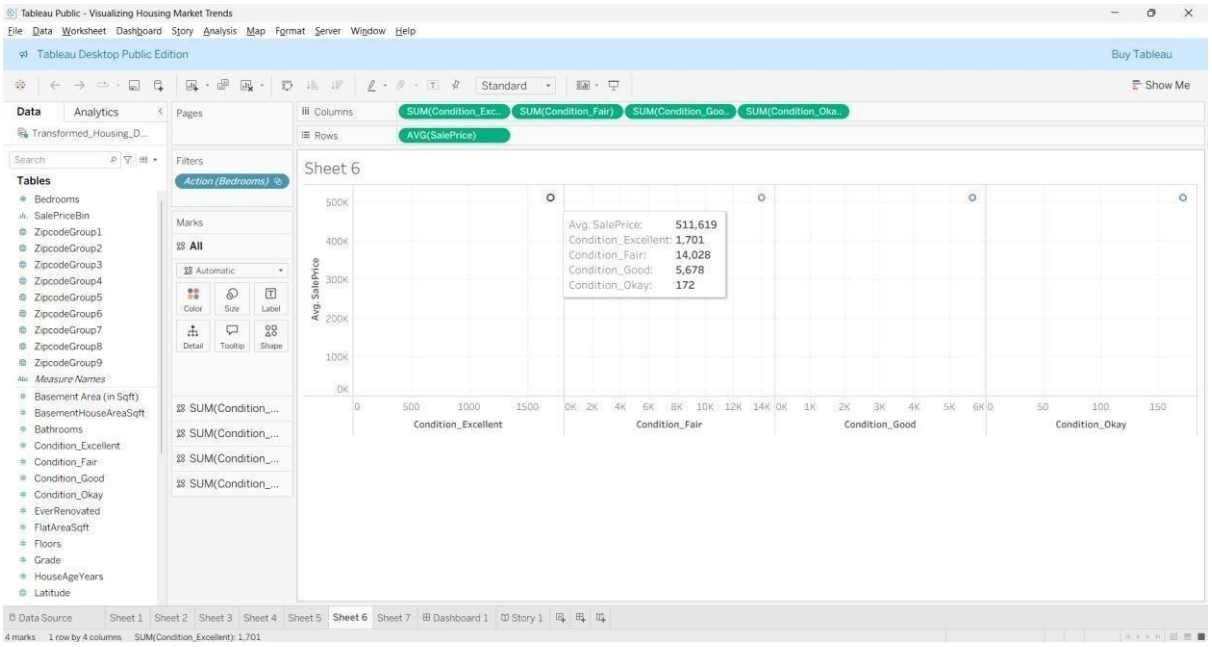
Output of

Sheet 5

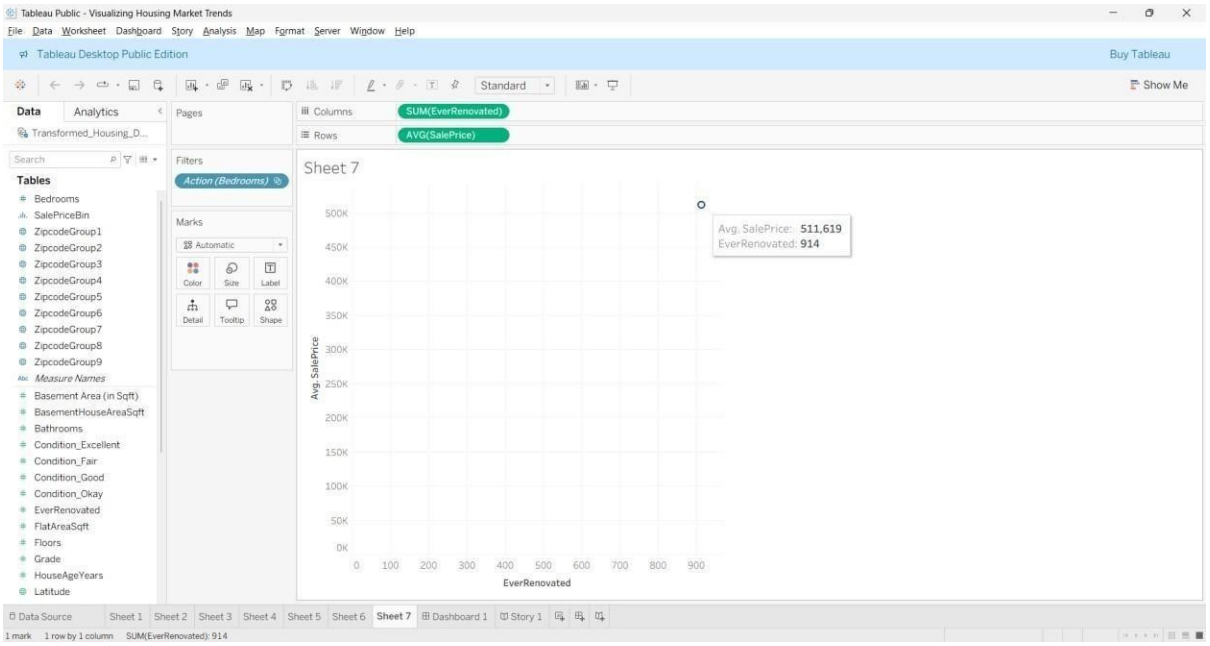


Output of Sheet 6

Output of

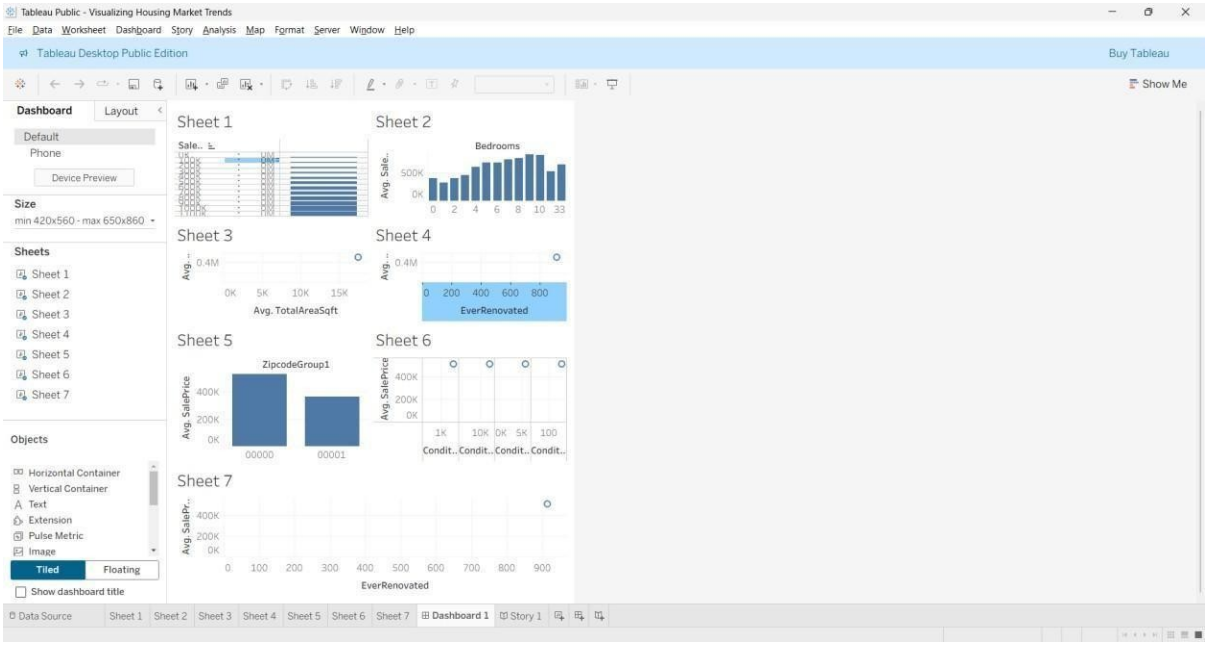


Sheet 7

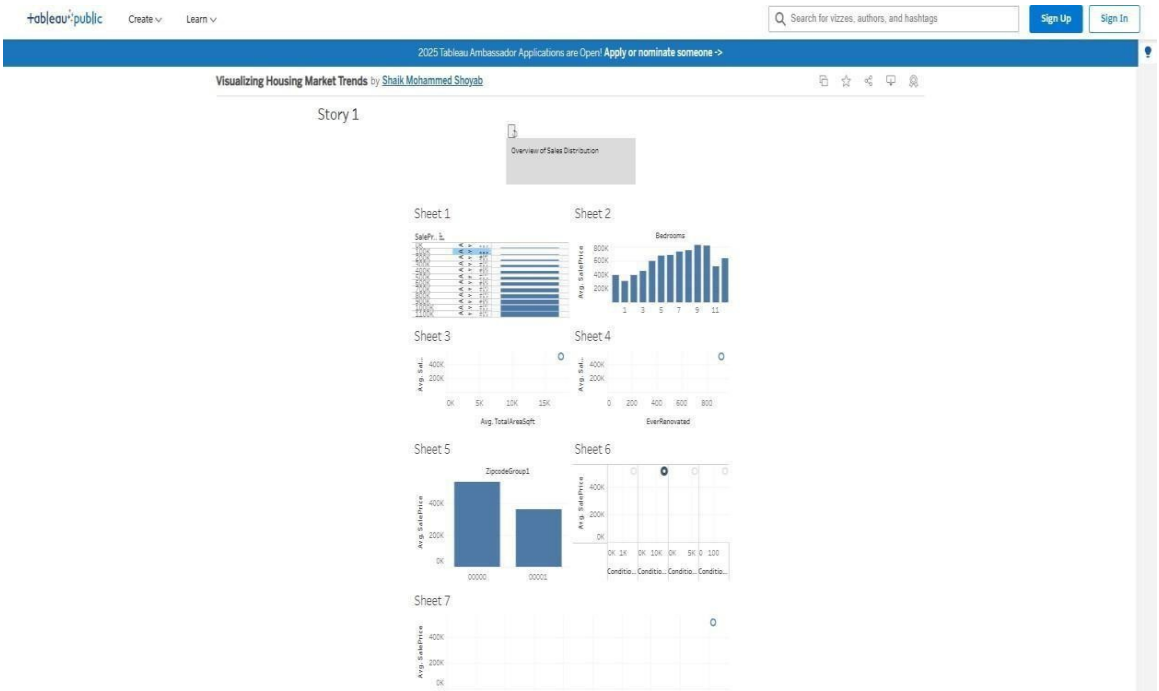


Output of

Output of Dashboard



Story



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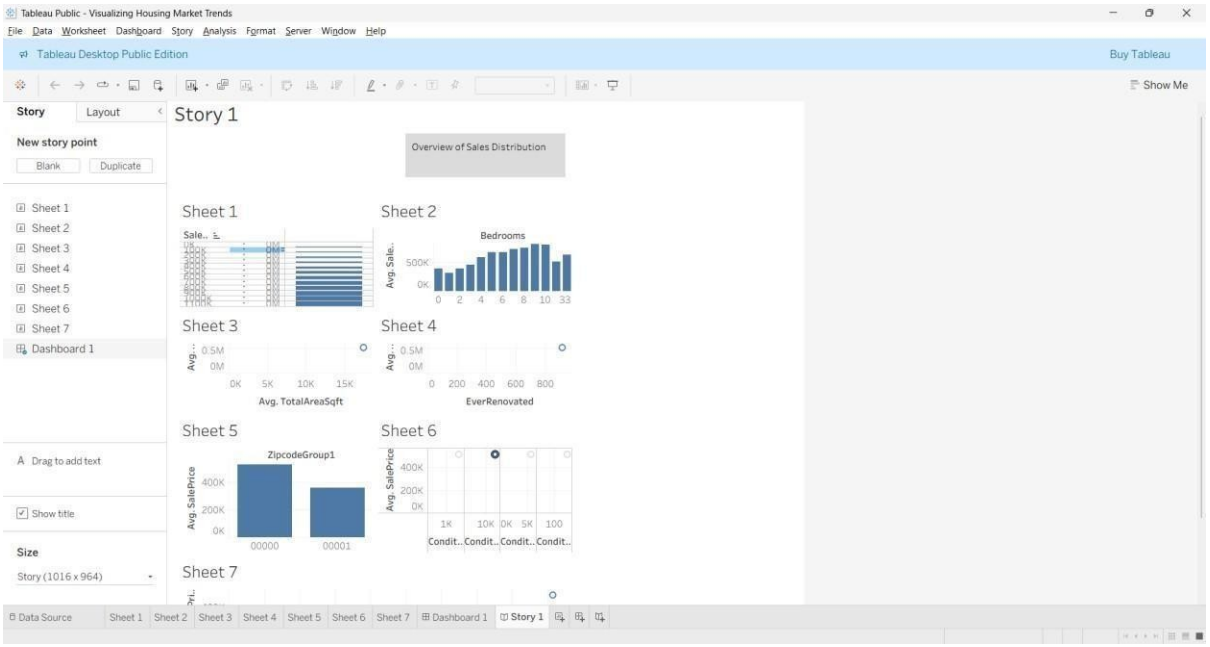

Tableau public link

https://public.tableau.com/app/profile/dhruv.sahu5081/viz/FInalSmartbridge/HouseAgeDistributionbyNumberofBathroomsBedroomsandFloors

Output

127.0.0.1:5000
Gmail
YouTube
Maps
News
Translate
All Bookmarks
Housing Market Trends Analysis
Sheet 1
Sheet 2
Bedrooms
Sheet 3
Avg. TotalAreaSqft
Sheet 4
EverRenovated
Sheet 5
ZipcodeGroup1
Sheet 6
Sheet 7
EverRenovated
View on Tableau Public
Share

8. Advantages & disadvantages

Advantages:

1. Interactive Analysis:

The Tableau dashboard allows users to explore data with filters (e.g., bedrooms, renovation status, price bins), enhancing understanding through dynamic interactions.

2. Informed Decision-Making:

Buyers, sellers, agents, and investors can make data-driven decisions by identifying which features (e.g., area, renovations, number of floors) impact property value.

3. Data Storytelling:

The Tableau Story feature presents insights in a sequential, digestible narrative—great for business reports or stakeholder presentations.

4. Geographic Visualization:

Zipcode grouping allows regional comparison of price trends and property types, revealing market opportunities and local disparities.

5. Calculated Metrics & KPIs:

Metrics like Average Sale Price and Total Area improve business clarity and enable fast comparisons across categories.

6. Web Accessibility:

Embedding the dashboard into a Flask web app increases accessibility—users can view it from any browser without needing Tableau Desktop.

7. Modular & Scalable Design:

The project structure supports additional data (e.g., rental prices, future years), making it expandable to other regions or market conditions.

8. Minimal Coding Required:

Most of the visualizations are created using Tableau's drag-and-drop interface—making it ideal for analysts without deep programming expertise.

Disadvantages:

1. Static Dataset Limitation:

The analysis depends on a preloaded CSV file; it doesn't support real-time updates unless integrated with live databases or APIs.

2. Tool Dependency:

The system relies on Tableau Public, which has limitations like no row-level security and requires dashboards to be public. **3. Learning Curve for Tableau:**

While Tableau is user-friendly, new users may need time to understand calculated fields, filters, and advanced charting options.

4. Limited Predictive Power:

This is a descriptive and visual analytics project—it does not use machine learning or predictive modeling to forecast housing prices.

5. Browser Compatibility:

Older browsers or low-resolution screens may not render complex dashboards optimally, especially if not designed responsively.

6. Manual Data Preprocessing:

Initial data cleaning, renaming, and transformation were done manually using Python or within Tableau, which might be error-prone at scale.

9. Conclusion:

The project "Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau" successfully demonstrates how complex real estate data can be transformed into meaningful, interactive visual insights. By leveraging Tableau's powerful visualization capabilities, we have made it easier for buyers, sellers, investors, and analysts to understand the key factors influencing house prices. Our dashboard enables quick comparisons based on features like number of bedrooms, renovations, house age, and geographic location. The integration with Flask provides a seamless web interface, enhancing accessibility and usability. Overall, this project bridges the gap between raw housing data and strategic real estate decision-making, allowing users to gain actionable insights with minimal technical expertise.

10. Future scope:

1. Live Data Integration:

Future versions can integrate live property listings or transaction data via APIs or real-time databases to provide up-to-date market insights.

2. Machine Learning Forecasting:

Incorporating regression models or time-series forecasting can help predict future housing prices based on historical trends and features.

3. Rental Market Visualization:

Extend the dashboard to include rental data analysis, enabling a broader comparison between buying vs. renting decisions.

4. Mobile Optimization:

Responsive design enhancements can be implemented to ensure the dashboard performs well across tablets and smartphones. **5. Advanced User Access Control:**

By using Tableau Server or Tableau Online, dashboards can be secured with role-based access for different stakeholders.

6. Location Intelligence Enhancements:

Integration of geospatial data, satellite maps, or demographic overlays can improve location-based insights (e.g., school zones, crime rates).

7. Recommendation Engine:

Develop a recommendation system to suggest optimal property types using user-input filters.

11. Appendix

Source Code:

index.html

```
<!-- templates/index.html -->
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Housing Market Dashboard</title>
</head>
<body>
   <h1 style="text-align: center;">_^¡ · Housing Market Trends Analysis</h1>
   <div class='tableauPlaceholder' id='viz1750827980701' style='position: relative'>
      <noscript>
         <a href='#'>
            <img alt='Dashboard 1 '
src='https://public.tableau.com/static/images/Vi/VisualizingHousingMarketTrends_175082
78 225630/Dashboard1/1_rss.png' style='border: none' /> </a>
      </noscript>
      <object class='tableauViz' style='display:none;'>
         <param name='host_url' value='https%3A%2F%2Fpublic.tableau.com%2F' />
         <param name='embed_code_version' value='3' />
         <param name='site_root' value='' />
         <param name='name'
value='VisualizingHousingMarketTrends_17508278225630&#47;Dashboard1' />
         <param name='tabs' value='no' />
         <param name='toolbar' value='yes' />
         <param name='static_image'
value='https://public.tableau.com/static/images/Vi/VisualizingHousingMarketTrends_17508
2 78225630/Dashboard1/1.png' />
         <param name='animate_transition' value='yes' />
         <param name='display_static_image' value='yes' />
         <param name='display_spinner' value='yes' />
```

```
        <param name='display_overlay' value='yes' />
        <param name='display_count' value='yes' />
        <param name='language' value='en-US' />
        <param name='filter' value='publish=yes' />
      </object>
    </div>
    <script type='text/javascript'> var divElement =
      document.getElementById('viz1750827980701'); var vizElement =
      divElement.getElementsByTagName('object')[0]; if
      (divElement.offsetWidth > 800) {
        vizElement.style.minWidth = '420px';
        vizElement.style.maxWidth = '650px'; vizElement.style.width =
        '100%'; vizElement.style.minHeight = '587px';
        vizElement.style.maxHeight = '887px'; vizElement.style.height =
        (divElement.offsetWidth * 0.75) + 'px';
      } else if (divElement.offsetWidth > 500) {
        vizElement.style.minWidth = '420px';
        vizElement.style.maxWidth = '650px'; vizElement.style.width =
        '100%'; vizElement.style.minHeight = '587px';
        vizElement.style.maxHeight = '887px'; vizElement.style.height =
        (divElement.offsetWidth * 0.75) + 'px';
      } else { vizElement.style.width =
        '100%'; vizElement.style.height =
        '1527px';
      } var scriptElement = document.createElement('script');
    scriptElement.src = 'https://public.tableau.com/javascripts/api/viz_v1.js';
    vizElement.parentNode.insertBefore(scriptElement, vizElement);
    </script>
  </body>
  </html>
```

app.py

```
from flask import Flask, render_template

app = Flask(_name_)
```

```
@app.route('/')
def home():
    return render_template('index.html')


if _name_____== '_main_':
    app.run(debug=True)
```

Project Structure

```
housing_dashboard/
├── app.py            # Flask server that renders the homepage
└── templates/
    └── index.html      # Web page embedding the Tableau dashboard
```

Dataset Link

https://docs.google.com/spreadsheets/d/1blBKrwunCQaiccy5sLP6mG4TsanJkO0C/edit?usp=drive_link&ouid=117818466889783119367&rtpof=true&sd=true

Project Demo Video Link

https://drive.google.com/file/d/1AreUu6c91xKUwJsU9KRdAL3mo_by4Nle/view?usp=drive_link

GitHub Repository Link

https://github.com/dhruv-41/smartbridgde